

Gokulraj M

Software Engineer · AI Engineer · Google Cloud / Looker Developer

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AI Engineer and Software Engineer with experience designing and building production-grade AI platforms, LLM applications, and cloud-native backend systems. Creator of Broks Forge, an AI Engineering Operating System that enables engineering teams to build, evaluate, debug, and operate AI applications through AI Git, deterministic evaluation, Engineering Intelligence, and Root Cause Explorer. Proficient in Java, Spring Boot, PostgreSQL, Docker, AWS, Google Cloud Platform (GCP), and modern Generative AI technologies.

Programming Languages	Java (Java 21), Javascript, C++, C#, Python, JavaScript, TypeScript, SQL
AI Engineering & Generative AI	Large Language Models (LLMs), Generative AI, Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agent Systems, AI Evaluation & Benchmarking, Prompt Versioning, AI Observability, Engineering Intelligence, Root Cause Analysis, Vector Search, Embeddings
Backend & Platform Engineering	Spring Boot, Spring Security, REST APIs, Microservices, PostgreSQL, Redis, Spring Data JPA, JWT Authentication, Multi-Tenant Architecture, AI Git, Deterministic Reasoning Systems
Frontend Development	React, Next.js, TypeScript, Tailwind CSS, HTML5, CSS3
Cloud, DevOps & Infrastructure	Google Cloud Platform (GCP), AWS EC2, Docker, Docker Compose, Nginx, GitHub Actions, CI/CD, Linux
AI Models & Development Tools	OpenAI, Anthropic Claude, Google Gemini, Groq, OpenRouter, ChromaDB, Cohere, Git, GitHub, Maven, Postman, Looker, LookML, IntelliJ IDEA, VS Code

EXPERIENCE

Tata Consultancy Services (TCS)

Dec 2025 – Present

Software Engineer, Google Cloud / Looker Developer

Chennai, Tamil Nadu

- Design and develop production-grade LookML models, views, and explores on Google Cloud Looker, enabling secure self-service analytics for Finance, Operations, and Marketing stakeholders.
- Restructured BigQuery SQL queries and aggregation logic, reducing average dashboard load time by ~35% across 10+ production Looker dashboards.
- Implemented row-level security and access-group policies in LookML to enforce multi-tenant data governance.

NIT Trichy

Sep 2024 – Oct 2024

Game Developer Intern

Trichy, Tamil Nadu

- Owned the end-to-end lifecycle of a game on online-bullying awareness, from design through deployment, using Unity and C#; optimized frame-rate and game-loop mechanics.

MIC Chennai

Aug 2023

VR Developer Intern

Chennai, Tamil Nadu

- Built immersive VR experiences with Unity's XR Interaction Toolkit and C#, improving simulation fidelity through iterative performance testing.

PRODUCTS

Brok's Forge — AI Engineering Operating System

broksforge.gokul.quest

Java 21, Spring Boot 3, PostgreSQL, Redis, Docker, AWS EC2, Next.js, TypeScript, GitHub Actions

- Architected and built Brok's Forge, an AI Engineering Operating System that provides a complete engineering workflow for building, evaluating, debugging, and operating production AI systems.
- Designed a provider-agnostic inference layer supporting OpenAI, Anthropic, Gemini, Groq, DeepSeek and OpenRouter through a unified abstraction independent of vendor SDKs.
- Built the Forge Kernel, a modular platform foundation powering AI Git, Registry, Brok, Execution Graph, Root Cause Explorer, Engineering Intelligence and evaluation workflows.
- Developed deterministic AI evaluation pipelines with immutable datasets, prompt versioning, benchmark execution, experiment tracking and reproducible results.
- Engineered Root Cause Explorer and Engineering Intelligence to analyze AI failures, trace execution history and surface actionable debugging insights across prompts, datasets and model versions.
- Implemented secure multi-tenant architecture with PostgreSQL, Redis, AES-256-GCM encrypted credentials, JWT authentication and tenant-isolated resource management.
- Designed and deployed the production platform on AWS EC2 using Docker, Nginx, TLS and automated GitHub Actions CI/CD with observability, health monitoring and zero-downtime deployment workflows.

Kuriosity — AI-Powered Student Learning Platform

kuriosity.gokul.quest

React, Vite, Node.js, Express, PostgreSQL, ChromaDB, Groq (LLM), Cohere (Embeddings), Resend, Railway, Vercel

- Built an AI-powered learning platform where students teach an AI student by uploading PDFs, PPTs and notes instead of chatting with a generic assistant.
- Engineered a production RAG pipeline using document parsing, chunking, Cohere embeddings and ChromaDB vector search to ground every response in uploaded learning material.
- Developed AI-based answer evaluation, topic mastery tracking and long-term learning memory to create personalized study sessions across multiple conversations.
- Implemented authentication, document processing, rate limiting and production deployment across Railway and Vercel with JWT and OTP authentication.

EDUCATION

SRM Institute of Science & Technology

Trichy, Tamil Nadu

B.Tech – Computer Science Engineering, CGPA: 7.83

May 2021 – Jul 2025

CERTIFICATIONS

- Associate Cloud Engineer (Google certified)
- HackerRank — Java (Certified)
- CodeChef — Python
- AR/VR Game Development — CDAC Pune

PROFILE LINKS

GitHub: github.com/gokulraj9488

Portfolio: gokul.quest

LinkedIn: linkedin.com/in/gokul-raj3003